Deep Learning for Economics and Finance

Overlapping generations, borrowing constraints, product-form KKT complementarity

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Companion notebooks: `07_OLG_Analytic_DEQN_exogenous.ipynb` / `08_OLG_Analytic_DEQN_persistent.ipynb` (analytic 6-agent), `09_OLG_Benchmark_DEQN_exogenous.ipynb` / `10_OLG_Benchmark_DEQN_persistent.ipynb` (benchmark 56-agent), and the student exercise notebook `11_OLG_Exercise.ipynb`.

Based on Azinovic, Gaegauf & Scheidegger (2022, *IER* 63(4), 1471–1525) and Krueger & Kübler (2004, *JEDC*).

Roadmap

Part I — What is an OLG model?

- Why overlapping generations; cohort structure; the DEQN recipe in one line.

Part II — The analytic 6-agent model (Krueger & Kübler, 2004)

- Model → equilibrium → cost → FREE → architecture → training → validation against the closed form.
- Primary notebook: `08_OLG_Analytic_DEQN_persistent.ipynb`; companion ablation `07_OLG_Analytic_DEQN_exogenous.ipynb`.

Part III — The leading example of AGS (2022): the 56-cohort benchmark

- Same 9-step recipe at research scale: two assets, two borrowing constraints, capital adjustment costs, persistent shocks, hump-shaped labor.
- Primary notebook: `10_OLG_Benchmark_DEQN_persistent.ipynb`; companion ablation `09_OLG_Benchmark_DEQN_exogenous.ipynb`.

Learning objectives

By the end of this lecture you should be able to:

- Construct a finite-dimensional overlapping-generations equilibrium with cohort-specific savings and lifecycle heterogeneity
- Derive equilibrium conditions and build the mean-squared Euler-residual loss for a multi-cohort OLG model
- Implement the DEQN recipe: model specification, equilibrium definition, cost function, network training, and validation
- Validate a learned solution against the closed-form equilibrium in the analytic Krueger–Kuebler model
- Scale the method from six cohorts to research benchmarks with capital adjustment costs, multiple assets, and binding constraints

Why Overlapping Generations?

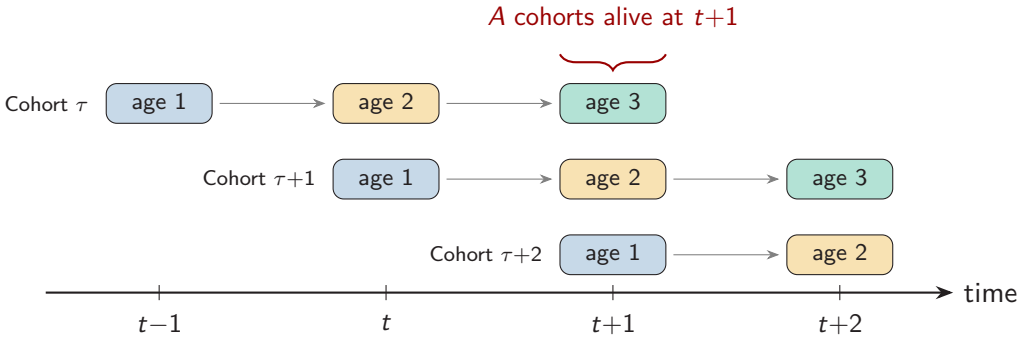
The economic content

- Lifecycle: agents are born, work, save, retire, die. A representative infinitely-lived household cannot speak to pensions, retirement, inheritance, or cohort-specific shocks.
- Demographics: at any t, many cohorts coexist, each with its own asset holdings and labor income.
- Finite asset distribution: with A cohorts, the cross-sectional wealth distribution is an A-vector – not an infinite-dimensional object. Perfect for DEQNs: state stays finite-dimensional with realistic heterogeneity.

Two canonical versions we solve today

1. $A = 6$ analytic model (Krueger & Kübler, 2004): closed form $\Rightarrow$ ground-truth validation.
2. $A = 56$ leading-example benchmark (Azinovic, Gaegauf & Scheidegger, 2022, §3): research-scale, no closed form.

OLG Structure: Overlapping Cohorts



At each period t we have A coexisting cohorts. A newborn enters (age 1); the oldest dies (age A).

Key observation: the number of types is finite (A), so the cross-sectional state is finite-dimensional.

The DEQN Recipe in One Line

From model to neural network (to be applied twice in today's lecture)

1. Write down the economic model: preferences, technology, constraints, shocks.
2. Define the equilibrium: choices + prices that satisfy optimization + market clearing.
3. Write the equilibrium conditions as a system $G(x, f(x)) = 0$.
4. Replace f by a neural network $\mathcal{N}_\theta$ and form the cost function $\mathcal{L}(\theta) = \frac{1}{|D|} \sum_j \|G(x_j, \mathcal{N}_\theta(x_j))\|^2$.
5. Sample training states D, minimize $\mathcal{L}(\theta)$ by SGD.

The only thing that depends on the model

The equilibrium conditions G. Everything else – the network, the SGD loop, the sampling method – is generic. So to solve a new model, write down its G.

Today's deliverable: derive G explicitly for both the analytic and the benchmark OLG models.

II.1 Setup — the Analytic Model

- Time discrete: $t = 0, 1, 2, \ldots, \infty$.
- Agents live for $A = 6$ periods.
- One representative household per cohort; a new one is born every period.
- No uncertainty about lifetime: all agents live exactly A periods.
- Exogenous aggregate shock $z_t \in \{1, 2, 3, 4\}$ drawn i.i.d. with transition $\pi_{ss'} = 0.25$.
- Log utility $u(c) = \ln c$ – chosen because it gives a closed-form equilibrium (Krueger & Kübler, 2004).
- Labor: only the youngest cohort works: $\ell^1 = 1,\ \ell^s = 0$ for $s \geq 2$.

Shocks: $z_t = (\eta_t, \delta_t)$

z	$\eta(z)$	$\delta(z)$	π
1	0.95	0.5	0.25
2	1.05	0.5	0.25
3	0.95	0.9	0.25
4	1.05	0.9	0.25

Calibration

$\alpha = 0.3,\ \beta = 0.7,\ \gamma = 1$ (log); $\underline{a} = 0$.

II.2 Households — Preferences and the Savings Technology

Agents supply their labor inelastically for a market wage w_t.

An agent currently of age s alive at t maximises remaining time-separable discounted expected lifetime utility ($\beta < 1$):

$$\mathbb{E}_t \left[\sum_{i=0}^{A-s} \beta^i \ \ln\left(c_{t+i}^{s+i}\right) \right] \qquad (1)$$

Households can save a unit of consumption to obtain a unit of capital good next period, denoted a_t^s.

The savings become capital in the next period:

$$a_t^s \ = \ k_{t+1}^{s+1}, \qquad \forall t, \ \forall s \in \{1, \ldots, A-1\}. \qquad (2)$$

II.3 Households — Constraints, Budget, Boundary

Households cannot die with debt. Borrowing is bounded:

$$a_t^s \;\geq\; \underline{a}. \qquad (3)$$

In the analytic model we set $\underline{a} = 0$: no borrowing for any cohort.

At time t, households sell their capital to the firm at market price $r_t > 0$. The per-period budget constraint of cohort s reads

$$c_t^s \;+\; a_t^s \;=\; r_t\, k_t^s \;+\; \ell^s\, w_t. \qquad (4)$$

Boundary conditions. Newborns enter with no inheritance, and the oldest cohort dies without saving anything:

$$k_t^1 \;=\; 0, \qquad\qquad a_t^A \;=\; 0.$$

II.4 Firms and Markets

There is a single representative firm with Cobb–Douglas production.

TFP η and depreciation δ depend on the exogenous shock z alone (see II.1 table).

Each period, after the shock has realized, the firm buys capital and hires labor to maximize profits, taking prices as given.

The stochastic production function is

$$f(K, L, z) \;=\; \eta(z)\, K^{\alpha}\, L^{1-\alpha} \;+\; K\big(1 - \delta(z)\big).$$

There are competitive spot markets for the consumption good, capital, and labor.

In the analytic model $L_t = \sum_{s=1}^{A} \ell^s = 1$ (only age-1 works), so the firm problem reduces to $\max_{K \geq 0} \eta(z) K^{\alpha} - (r_t - (1 - \delta(z)))K$.

II.5 Competitive Equilibrium — Definition

Definition 1 (competitive equilibrium)

Given initial conditions z_0, $\{k_0^s\}_{s=1}^{A-1}$, a competitive equilibrium is a collection of choices for households $\left\{(c_t^s,\ a_t^s)_{s=1}^{A}\right\}_{t=0}^{\infty}$, for the representative firm $(K_t, L_t)_{t=0}^{\infty}$, and prices $(r_t, w_t)_{t=0}^{\infty}$, such that:

1. Given $(r_t, w_t)_{t\geq 0}$, household choices maximize the lifetime utility (1) subject to the savings-to-capital identity (2), the borrowing limit (3), and the per-period budget (4):
   $$a_t^s = k_{t+1}^{s+1};\quad a_t^s \geq \underline{a};\quad c_t^s + a_t^s = r_t k_t^s + \ell^s w_t.$$
2. Given (r_t, w_t), the firm maximises profits:
   $$(K_t, L_t) \in \arg\max_{K,L\geq 0} f(K, L, z_t) - r_t K - w_t L.$$
3. All markets clear: for all t, $\quad L_t = \sum_{s=1}^{A} \ell^s,\quad K_t = \sum_{s=1}^{A} k_t^s.$

II.6 Equilibrium Conditions

Firm FOCs (closed form from Cobb–Douglas):

$$w_t \;=\; (1-\alpha)\,\eta(z_t)\,K_t^{\alpha}\,L_t^{-\alpha}, \qquad r_t \;=\; \alpha\,\eta(z_t)\,K_t^{\alpha-1}\,L_t^{1-\alpha} \;+\; \big(1-\delta(z_t)\big).$$

Household optimality for any cohort of age $s \in \{1,\dots,A-1\}$ (FOC in c_t^s gives $\nu_t^s = u'(c_t^s)$; FOC in a_t^s gives the standard interior Euler equation):

$$\boxed{\underbrace{u'\big(c_t^{\,s}\big)}_{=1/c_t^s \text{ since } u=\ln} \;=\; \beta\,\mathbb{E}\big[u'\big(c_{t+1}^{\,s+1}\big)\,r_{t+1}\big]}$$

Terminal cohort. Age A: consume everything, save nothing.

Why no KKT multiplier? In this calibration ($\gamma = 1$, $\beta = 0.7$, only the youngest cohort works) the borrowing constraint $a_t^s \geq 0$ never binds in equilibrium, so $\lambda_t^s \equiv 0$ and drops out of the Euler equation, the FREE signature, and the loss. Part III re-introduces multipliers for the binding constraints of the 56-cohort benchmark.

II.7 The DEQN Cost Function

Explicit cost function

$$\boxed{\mathcal{L}_{D_{\text{train}}}(\theta) \;:=\; \frac{1}{|D_{\text{train}}|}\,\frac{1}{A-1}\sum_{x_j\in D_{\text{train}}}\;\sum_{i=1}^{A-1}\big(e^{i}_{\text{REE}}(x_j)\big)^2}$$

Relative Euler-equation residual (cohort i):

$$e^{i}_{\text{REE}}(x_j) \;:=\; \frac{\left(u'\right)^{-1}\!\left(\beta\,\mathbb{E}\big[r(\hat{x}_{j,+})\;u'\big(\hat{c}^{\,i+1}(\hat{x}_{j,+})\big)\big]\right)}{\hat{c}^{\,i}(x_j)} \;-\; 1.$$

Sampling next state. $\hat{x}_{j,+} = \big(z_+,\ 0,\ \hat{a}^{\,[1:A-1]}(x_j)\big)$ with z_+ drawn from π (newborn enters with $k^1 = 0$).

Feasibility. $a^s \geq 0$ and $c^s \geq 0$ are enforced by the sigmoid savings-fraction output of II.9 (each $a^s \in [0, \mathrm{inc}^s]$); small soft penalties on negative c and K are added in the notebook as numerical backstops but are inactive with this head.

II.8 Functional Rational Expectations Equilibrium (FREE)

FREE: one function that solves the equilibrium conditions at every state

Signature: $f : \{1,2,3,4\} \times \mathbb{R}^A \longrightarrow \mathbb{R}^{A-1}$, input = state $(z_t, \boldsymbol{k}_t)$, output = savings policies.

Output decomposition ($A-1$ scalars):

$$f\left(\begin{bmatrix} z_t \\ \boldsymbol{k}_t \end{bmatrix}\right) = \Big[\underbrace{a_t^1,\ a_t^2,\ \ldots,\ a_t^{A-1}}_{\text{capital-investment policies}}\Big]^\top.$$

Such that, for all cohorts $s = 1, \ldots, A-1$, the (interior) Euler equation holds:

$$\beta\,\mathbb{E}\left[\frac{r_{t+1}\,u'(c_{t+1}^{s+1})}{u'(c_t^s)}\right] - 1 = 0 \qquad \text{(Euler residual).}$$

Stack these $A-1$ residuals as $G\big([z_t, \boldsymbol{k}_t],\ f([z_t, \boldsymbol{k}_t])\big) = \mathbf{0}$.

$A = 6 \Rightarrow 5$ outputs. Notebook uses an extended state of dim $16 + 4A = 40$ (pre-computed aggregates + per-agent income blocks + the transition-probability row) for ease of learning – pure feature engineering.

II.9 Deep Equilibrium Net — Architecture (Analytic)

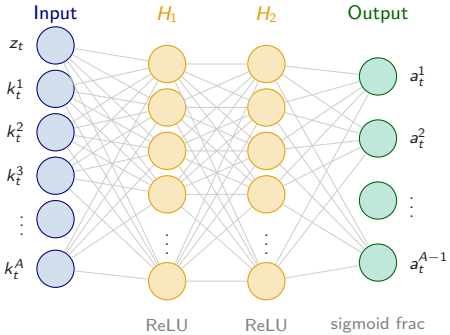
$$\mathcal{N}_\theta : \{1,2,3,4\} \times \mathbb{R}^A \to \mathbb{R}^{A-1}$$

$$\mathcal{N}_\theta\left(\begin{bmatrix} z_t \\ \boldsymbol{k}_t \end{bmatrix}\right) = \begin{bmatrix} a_t^1 \\ a_t^2 \\ \vdots \\ a_t^{A-1} \end{bmatrix}$$

such that

$$G\left(\begin{bmatrix} z_t \\ \boldsymbol{k}_t \end{bmatrix}, \mathcal{N}_\theta\left(\begin{bmatrix} z_t \\ \boldsymbol{k}_t \end{bmatrix}\right)\right) \approx \mathbf{0}.$$

Sigmoid savings-fraction output $\Rightarrow$ $0 \le a_t^s \le \mathrm{inc}_t^s$ by construction, so $a_t^s \ge 0$ (borrowing constraint) and $c_t^s \ge 0$ both hold at every iteration without an explicit multiplier.



2 hidden layers $\times$ (100, 50) units; single forward pass.

Companion Notebook Hand-over

Open notebook `08_OLG_Analytic_DEQN_persistent.ipynb` (the persistent-simulation primary; the exogenous-sampling companion is `07_OLG_Analytic_DEQN_exogenous.ipynb`). It implements the model, equilibrium conditions, loss, FREE map, architecture, training, and validation covered in Part II.

Cell-to-slide map

Deck slide	Notebook section
II.1–II.6 Model and equilibrium	§1–3 parameters, state, network/policy
II.7–II.9 Loss, FREE, architecture	§4 `compute_cost()` + training-data construction
II.10–II.13 Training/validation	§5–8 mini-batches, diagnostics, policy stability, closed-form checks

II.10 Sampling and the Training Loop

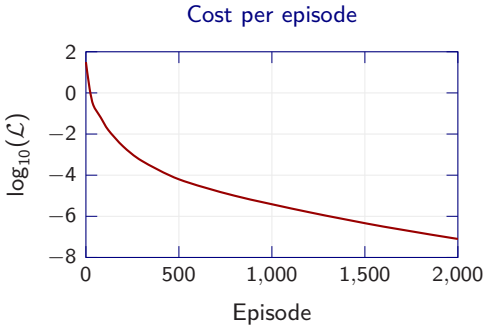
Cloud method (Azinovic, Gaegauf & Scheidegger, 2022)

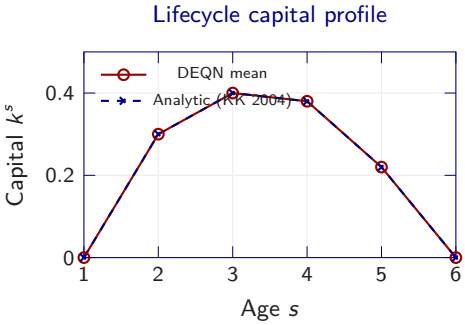
1. Initialisation. Sample a cloud of initial states $D_0 = \{x_j^0\}_{j=1}^N$; initialise θ^0.
2. Episode k.
   (a) **Simulate.** Draw fresh shocks $\{z_{+,j}\}$ from π; roll the cloud forward one period using the current policy $\mathcal{N}_{\theta^k}$: $x_j^{k+1} = \left(z_{+,j},\, 0,\, \mathcal{N}_{\theta^k}^{[1:A-1]}(x_j^k)\right)$.
   (b) **Mini-batch SGD.** Take steps Adam steps on $\mathcal{L}_{D_k}(\theta)$ using mini-batches from D_k.
3. Stop when $\log_{10} \mathcal{L}$ falls below the target.

Definitions. Episode = one simulate-then-train pass. Epoch = one pass through the full cloud.

Sampling from the ergodic set of the current policy makes the cost function focus on states the economy actually visits.

II.11 Training Results — Analytic Model





Reading the plots

Left: cost falls in $\log_{10}$ as the network learns the Krueger–Kübler policy; teaching is diagnostic, production reaches tight residuals.
Right: trained DEQN savings policy overlays the closed-form Krueger–Kübler solution across cohorts.

Finger Exercise: Closed-Form Savings Rates

Setup. With $u = \ln$, cohort h's Euler equation reads $1/c_t^h = \beta\,\mathbb{E}_t[r_{t+1}/c_{t+1}^{h+1}]$ and the budget constraint is $c_t^h + a_t^h = \text{inc}_t^h$ where $\text{inc}_t^h = r_t k_t^h + \ell^h w_t$. (The household-level setup slides II.2–II.6 used the index symbol s; from this slide onward we follow the script convention and use h instead.)

Claim (closed-form log-utility OLG savings rate; the validation target used by Krueger & Kübler, 2004). With log utility and 100% consumption at age A, the savings rate $\beta_h := a_t^h/\text{inc}_t^h$ of cohort h depends only on age:

$$\beta_h \;=\; \beta\,\frac{1-\beta^{A-h}}{1-\beta^{A-h+1}}, \qquad h = 1, \ldots, A-1.$$

Task (3 minutes). Compute $\beta_1, \beta_2, \ldots, \beta_5$ with $\beta = 0.7$. Interpret:

- What happens as cohorts approach retirement?
- Why are younger cohorts saving more as a rate?

Solution: Closed-Form Savings Rates

$$\beta_h = 0.7 \cdot \frac{1 - 0.7^{6-h}}{1 - 0.7^{7-h}}$$

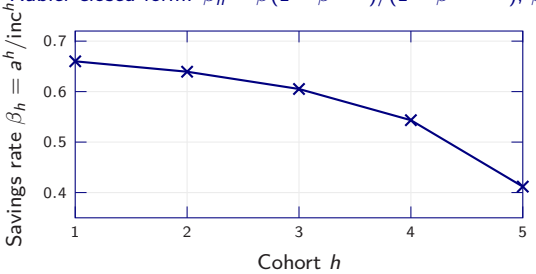
Cohort h	Periods remaining $(A - h)$	Savings rate β_h
1	5	$0.7 \cdot (1 - 0.7^5)/(1 - 0.7^6) \approx 0.660$
2	4	$0.7 \cdot (1 - 0.7^4)/(1 - 0.7^5) \approx 0.639$
3	3	$0.7 \cdot (1 - 0.7^3)/(1 - 0.7^4) \approx 0.605$
4	2	$0.7 \cdot (1 - 0.7^2)/(1 - 0.7^3) \approx 0.543$
5	1	$0.7 \cdot (1 - 0.7)/(1 - 0.7^2) \approx 0.412$
6	0	0 (consumes everything)

Interpretation

- Savings rate monotonically declines with age: as the horizon shrinks, saving less is optimal.
- Youngest cohort saves $\sim 66\%$ of its income – longest horizon to enjoy the returns.
- At A: terminal boundary $a^A = 0$ – consume everything.

II.13 DEQN vs. Analytical Solution

Krueger–Kübler closed form: $\beta_h = \beta(1-\beta^{A-h})/(1-\beta^{A-h+1})$, $\beta = 0.7$, $A = 6$



Validation strategy

The plotted points are the exact closed form from the finger-exercise solution slide. The DEQN should reproduce this curve cohort-by-cohort.

The actual training run, DEQN-vs-analytic comparison, and Euler-error histogram are produced by `08_OLG_Analytic_DEQN_persistent.ipynb` (§8). Production target in AGS (2022): mean relative Euler error $\lesssim 10^{-3}$.

II Take-away and Hands-on

What we just did

1. OLG model with 6 cohorts, log utility, 4 i.i.d. shocks (II.1–II.4).
2. Equilibrium definition + conditions (II.5–II.6).
3. Conditions $\rightarrow$ cost function (II.7).
4. Equilibrium map $f \rightarrow$ neural network $\mathcal{N}_\theta$ (II.8–II.9).
5. Trained by sampling + SGD (II.10–II.11), validated against the closed form (II.13).

Hands-on

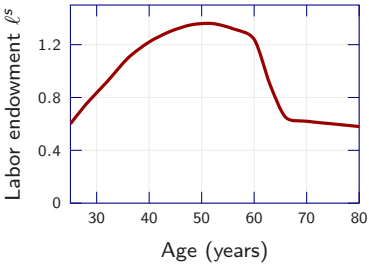
Open `08_OLG_Analytic_DEQN_persistent.ipynb`. Set `RUN_MODE = "teaching"` ($\sim$ 5 min) for the classroom run; `"smoke"` (the default) is a $\sim$30-s sanity check; `"production"` chases AGS Table 3 accuracy. Both notebooks expose a `SAMPLING_MODE = "simulation" / "exogenous"` switch (the persistent primary uses `"simulation"`, the companion `07_OLG_Analytic_DEQN_exogenous.ipynb` uses `"exogenous"`); the model and loss are identical – only the training cloud differs.

III.1 Setup — the Leading Example of AGS (2022)

Same recipe as Part II. What changes:

- Agents live for $A = 56$ periods (ages 25–80); one period = one year.
- CRRA utility $u(c) = c^{1-\gamma}/(1-\gamma)$ with $\gamma = 2$ (not log).
- Hump-shaped labor endowment ℓ^s (Kotlikoff-style; right panel).
- Persistent Markov shocks on (η, δ) (not i.i.d.).
- Two assets: capital k^s and bonds b^s.
- Two constraints: borrowing $k'^s \geq 0$ and collateral $k'^s + \kappa b'^s \geq 0$.
- Capital adjustment costs $\Psi^s = \frac{\zeta}{2}(k'^s - rk^s)^2$.

Stylised lifecycle labor (schematic)



Schematic shape only, rises through working years, drops sharply at retirement, low but nonzero after. See Brumm, Kübler & Scheidegger (2017) for the exact ℓ^s used in the notebook; AGS (2022) reuse this calibration as part of their benchmark.

III.2 Households — Preferences and Savings Technology

An agent of age s alive at t maximises

$$\mathbb{E}_t\left[\sum_{i=0}^{A-s}\beta^i\, u\big(c_{t+i}^{s+i}\big)\right], \qquad u(c)=\frac{c^{1-\gamma}}{1-\gamma}, \qquad \gamma=2. \qquad (1')$$

Two saving instruments (instead of one):

- Capital k_{t+1}^{s+1}: yields gross return r_{t+1} next period; incurs an adjustment cost $\Psi_t^s=\frac{\zeta}{2}(k_{t+1}^{s+1}-r_t k_t^s)^2$.
- Bonds b_{t+1}^{s+1}: traded at price p_t today, pay one unit of consumption next period. Zero net supply: $\sum_{s=1}^{A} b_t^s=0$.

Savings become next-period asset holdings ($\forall s\in\{1,\dots,A-1\}$):

$$k_{t+1}^{s+1} \;=\; a_{t,k}^s, \qquad b_{t+1}^{s+1} \;=\; a_{t,b}^s. \qquad (2')$$

III.3 Households — Constraints, Budget, Boundary

Two constraints on the capital–bonds pair (k'^s, b'^s)**:**

$$k'^s \;\geq\; 0 \quad \text{(borrowing constraint)}, \qquad\qquad k'^s \;+\; \kappa\, b'^s \;\geq\; 0 \quad \text{(collateral constraint)}. \qquad (3')$$

Budget constraint of cohort s (bonds enter at price p_t on the left-hand side, pay off on the right):

$$c_t^s \;+\; k_{t+1}^{s+1} \;+\; p_t\, b_{t+1}^{s+1} \;+\; \Psi_t^s \;=\; r_t\, k_t^s \;+\; b_t^s \;+\; \ell_t^s\, w_t. \qquad (4')$$

Boundary conditions (born and die without assets):

$$k_t^1 = 0,\;\; b_t^1 = 0; \qquad\qquad k_{t+1}^{A+1} = 0,\;\; b_{t+1}^{A+1} = 0.$$

Bond parameter $\kappa \geq 0$ governs how much bond-short position capital collateralises; $\kappa = 0$ is a hard short-sale constraint on bonds.

III.4 Firms and Markets — Adding the Bond Market

Representative firm. Same Cobb–Douglas as Part II:

$$f(K, L, z) \;=\; \eta(z)\, K^{\alpha}\, L^{1-\alpha} \;+\; K\big(1 - \delta(z)\big).$$

Labor now aggregates over the hump-shaped profile: $L_t = \sum_{s=1}^{A} \ell^s$ (a constant).

Markets cleared in equilibrium.

1. Labor: $L_t = \sum_{s=1}^{A} \ell^s$.
2. Capital: $K_t = \sum_{s=1}^{A} k_t^s$.
3. Bonds (new): $0 \;=\; \displaystyle\sum_{s=1}^{A} b_t^s$. (Zero net supply.)

Aggregate shocks. $z_t \in \{1, \ldots, 4\}$ as before, but transition π^z is persistent (not i.i.d.): (η, δ) have their own Markov chains, tensor-producted.

III.5 Competitive Equilibrium — Definition

Definition 1' (competitive equilibrium, benchmark)

Given initial conditions z_0, $\{k_0^s, b_0^s\}_{s=1}^{A-1}$, a competitive equilibrium is a collection of choices $\left\{(c_t^s,\ k_t^{s+1},\ b_t^{s+1})_{s=1}^{A-1}\right\}_{t=0}^{\infty}$, firm demands $(K_t, L_t)_{t\geq 0}$, and prices $(r_t, w_t, p_t)_{t\geq 0}$, such that:

1. Households maximize $(1')$ subject to $(2')$–$(4')$.
2. Firm maximises profits, same closed-form FOCs as Part II.
3. All three markets clear every period: capital, labor, bonds.

Relative to Part II: one new price (p_t), one new market-clearing condition (bonds), one new asset class per cohort in the choice vector. Everything else is the same.

III.6 Equilibrium Conditions (Benchmark)

Firm FOCs. Same closed form as Part II.

Two Euler equations per cohort, for $s = 1, \ldots, A-1$:

$$\text{(capital)} \;\; u'(c_t^s) \cdot \big(1 + \zeta\,(k'^s - r_t k_t^s)\big) \;=\; \beta\,\mathbb{E}\big[u'(c_{t+1}^{s+1})\, r_{t+1}\big] \;+\; \lambda_{\mathrm{b}}^s \;+\; \lambda_{\mathrm{c}}^s,$$

$$\text{(bonds)} \;\; u'(c_t^s) \cdot p_t \;=\; \beta\,\mathbb{E}\big[u'(c_{t+1}^{s+1})\big] \;+\; \kappa\,\lambda_{\mathrm{c}}^s.$$

Two KKT triples per cohort.

$$\lambda_{\mathrm{b}}^s \geq 0, \quad k'^s \geq 0, \quad \lambda_{\mathrm{b}}^s \cdot k'^s = 0; \qquad \lambda_{\mathrm{c}}^s \geq 0, \quad k'^s + \kappa b'^s \geq 0, \quad \lambda_{\mathrm{c}}^s \cdot (k'^s + \kappa b'^s) = 0.$$

Terminal cohort. Age A: consume everything.

So per state we have $4(A-1)$ per-agent residuals $+$ 1 bond market-clearing residual $= \mathbf{4\,(A-1)+1}$ equations.

III.7 The DEQN Cost Function (Benchmark)

Five loss components, one cost function

$$\mathcal{L}_{D_{\text{train}}}(\theta) \;=\; \frac{1}{|D_{\text{train}}|}\,\frac{1}{4(A-1)+1}\sum_{x_j\in D_{\text{train}}}\left[\sum_{s=1}^{A-1} R^{s}(x_j)^2 \;+\; \left(e_{\text{MC},b}(x_j)\right)^2\right],$$

where $R^{s}(x_j)^2 \;:=\; \left(e^{s}_{\text{REE},k}\right)^2 + \left(e^{s}_{\text{REE},b}\right)^2 + \left(e^{s}_{\text{KKT},b}\right)^2 + \left(e^{s}_{\text{KKT},c}\right)^2.$

Component map.

- $e^{s}_{\text{REE},k}$, $e^{s}_{\text{REE},b}$: relative residuals of the capital and bond Euler equations.
- $e^{s}_{\text{KKT},b} = \lambda^{s}_{\text{b}} \cdot k'^{s}$: borrowing KKT.
- $e^{s}_{\text{KKT},c} = \lambda^{s}_{\text{c}} \cdot (k'^{s} + \kappa b'^{s})$: collateral KKT.
- $e_{\text{MC},b} = \sum_{s=1}^{A} b_t^{s}$: bond market clearing.

With $A = 56$ this is $4 \times 55 + 1 = 221$ loss terms per training state. The structure is the same as Part II: squared equilibrium residuals.

III.8 Functional Rational Expectations Equilibrium

FREE for the benchmark

$$f : \{1, \ldots, 4\} \times \mathbb{R}^A \times \mathbb{R}^A \longrightarrow \mathbb{R}^{4(A-1)+1}$$

$$f\left(\begin{bmatrix} z_t \\ \boldsymbol{k}_t \\ \boldsymbol{b}_t \end{bmatrix}\right) = \begin{bmatrix} \hat{k}_t'^1, \ldots, \hat{k}_t'^{A-1} \\ \hat{\lambda}_{b,t}^1, \ldots, \hat{\lambda}_{b,t}^{A-1} \\ \hat{q}_t^1, \ldots, \hat{q}_t^{A-1} \\ \hat{\mu}_t^1, \ldots, \hat{\mu}_t^{A-1} \\ \hat{p}_t \end{bmatrix}^\top$$

Output blocks: capital savings; borrowing multipliers; collateral $\hat{q}^h \equiv \hat{k}'^h + \kappa\, \hat{b}'^h$; collateral multipliers; equilibrium bond price (one scalar).

Such that $G\big([z_t, \boldsymbol{k}_t, \boldsymbol{b}_t],\ f([z_t, \boldsymbol{k}_t, \boldsymbol{b}_t])\big) = \mathbf{0}$ where G stacks the $4(A-1)+1$ equations of III.6.

With $A = 56$: minimal state dim $= 113$; the notebook input is 240 after engineered aggregates and transition probabilities. Output dim $= 4 \times 55 + 1 = \mathbf{221}$. Bond holdings are recovered as $\hat{b}'^h = (\hat{q}^h - \hat{k}'^h)/\kappa$, and softplus enforces $\hat{k}'^h, \hat{\lambda}_b^h, \hat{q}^h, \hat{\mu}^h \geq 0$.

III.9 Architecture — Benchmark

$$\mathcal{N}_\theta : \mathbb{R}^{240} \to \mathbb{R}^{221}$$

Sizes from `10_OLG_Benchmark_DEQN_persistent.ipynb`

Notebook input	240 (minimal state: 113)
Hidden layers	2×128 (teaching)
	2×1000 (production)
Output dim	$4(A-1) + 1 = 221$
Output layer	softplus on every output
Optimizer	Adam, lr 10^{-5}

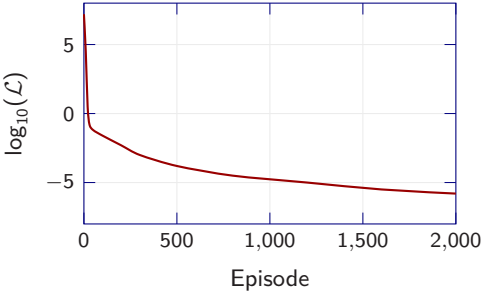
Output slicing

$$\mathcal{N}_\theta(x_t) = \left(\hat{k}'^{1:A-1},\ \hat{\lambda}_b^{1:A-1},\ \hat{q}^{1:A-1},\ \hat{\mu}^{1:A-1},\ \hat{p}\right).$$

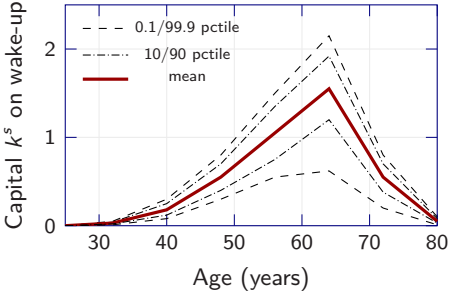
Same MLP (template) as Part II, scaled up and augmented by a bond-price head. Bond holdings are recovered from $\hat{k}'^{1:A-1}$, $\hat{q}^{1:A-1}$, $\hat{\mu}^{1:A-1}$.

III.10 Training — Loss and Lifecycle Capital

Cost per episode (teaching, 128^2, 2,000 ep.)



Lifecycle capital



Reading the plots

Left: loss falls sharply but oscillates because each episode re-simulates the cloud.
Right: lifecycle capital is hump-shaped: accumulation during working years, decumulation after retirement. Bands are simulated percentiles.

III.11 Consumption Profiles by Aggregate State

What the notebook will plot

For each aggregate state $z \in \{1, 2, 3, 4\}$ we obtain a lifecycle consumption profile $c^s(z)$ over ages $s = 25, \ldots, 80$, evaluated on a long ergodic simulation from `10_OLG_Benchmark_DEQN_persistent.ipynb`.

The four aggregate states

z	$\eta(z)$	$\delta(z)$	expected consumption ranking
2	1.022	0.08	highest – best technology, low depreciation
4	1.022	0.11	high
1	0.978	0.08	low
3	0.978	0.11	lowest – bad technology, high depreciation

What to look for in the live plot

Four distinct profiles, ordered as in the table; exact magnitudes and crossover ages come from §11 of the notebook.

III.12 Reading the Diagnostics

How do we know the network actually solved the equilibrium?

1. Loss level. $\log_{10} \mathcal{L} \lesssim -5$ means equilibrium residuals at the 10^{-3} level or smaller – AGS (2022) threshold.
2. Per-equation Euler errors. Plot mean $|e^s_{\mathrm{REE},k}|$ and $|e^s_{\mathrm{REE},b}|$ by age s; expect $\sim 10^{-3}$ everywhere.
3. Bond market clearing. $|e_{\mathrm{MC},b}| < 10^{-4}$ on the ergodic simulation path.
4. Borrow-binding fraction. Ages where $\lambda^s_{\mathrm{b}} > 0$; standard calibration: only the youngest few cohorts.
5. Policy-drift on a fixed anchor cloud. Evaluate the policy on `X_anchor` each monitoring interval; report `policy_drift_rms`, `policy_drift_max`; flag the run *time-invariant* once both fall below `TIME_INVARIANCE_TOL_RMS` / `_MAX`.
6. Ergodic simulation. Simulate 10 000 periods; plot lifecycle consumption + capital with percentile bands.

Hands-on: `10_OLG_Benchmark_DEQN_persistent.ipynb` (persistent-simulation primary; `09_OLG_Benchmark_DEQN_exogenous.ipynb` is the feedback-free ablation, same model and loss, different training cloud). §1–4 match III.1–III.5; §5–7 = training and policy stability; §8 = plots and diagnostics of III.10–III.12. Teaching runtime (128-wide, 2 000 episodes): $\approx$ 30 min on CPU.

Summary

What we accomplished

1. Introduced OLG as a finite-dimensional family of heterogeneous-agent models – well suited to DEQNs because the aggregate state stays finite-dimensional.
2. Applied the DEQN recipe twice:
   - Part II: the 6-agent Krueger–Kübler (2004) analytic model, validated against the closed form ($\lesssim 10^{-3}$ mean Euler error).
   - Part III: the 56-cohort leading-example benchmark of AGS (2022): CRRA, hump labor, persistent shocks, two assets, two borrowing constraints, capital adjustment costs.
3. Same 9-step recipe both times: setup, households I/II, firms, equilibrium, conditions, cost, FREE, architecture.

One line to remember

Writing down a DEQN = writing down the equilibrium conditions G. Everything else (network, SGD, sampling) is boilerplate.

References and Further Reading

- Azinovic, M., Gaegauf, L., & Scheidegger, S. (2022). *Deep equilibrium nets.* International Economic Review 63(4), 1471–1525. [notebook 10 (`benchmark_persistent`) implements §3 of this paper].
- Krueger, D., & Kübler, F. (2004). *Computing equilibrium in OLG models with stochastic production.* Journal of Economic Dynamics & Control 28(7), 1411–1436. [notebook 08 (`analytic_persistent`) implements their 6-agent analytic model].
- Fischer, A. (1992). *A special Newton-type optimization method.* Optimization 24, 269–284. [FB-complementarity function – used in the **IRBC chapter (Ch. 3)**; mentioned in this lecture only as an alternative to the product-form KKT residuals adopted here.]
- Code: https://github.com/sischei/DeepEquilibriumNets.

Optional Next: Agentic-Programming Workshop

What's next (optional, no lecture-script chapter). A hands-on workshop in which an LLM-driven coding agent (Claude Code, Cursor, Codex) acts as a pair programmer on a fresh DEQN model.

Self-study extension (notebooks 10–12, deck 08). Heterogeneous agents on a histogram (Young's method), continuum-of-agents DEQN, and Krusell–Smith with deep learning – visit any time before the PINN material.